DATA QUALITY MANAGEMENT FRAMEWORK

Enterprise Data Governance Framework Design & Implementation for Digital Banking

Framework ID	:	DG-DQF-001
Framework Owner	:	Chief Data Officer (CDO)
Approved By	:	Data Governance Council
Review Cycle	:	Annual
Framework Version	:	1.0

1. EXECUTIVE SUMMARY

The Data Quality Management Framework (DQMF) establishes the governance, processes, controls, standards, roles, and metrics required to ensure that data used across the Digital Bank is accurate, complete, consistent, timely, valid, and unique.

The framework supports:

- Regulatory Compliance
- BCBS 239 Compliance
- Risk Management
- Customer Experience
- Digital Banking Operations
- Analytics & AI Initiatives
- Enterprise Data Governance Objectives

2. FRAMEWORK OBJECTIVES

The Digital Bank shall implement a Data Quality Management Framework to:

✓ Improve trust in enterprise data

✓ Reduce operational risk

✓ Improve regulatory reporting accuracy

✓ Support customer-centric decision making

✓ Reduce data-related incidents

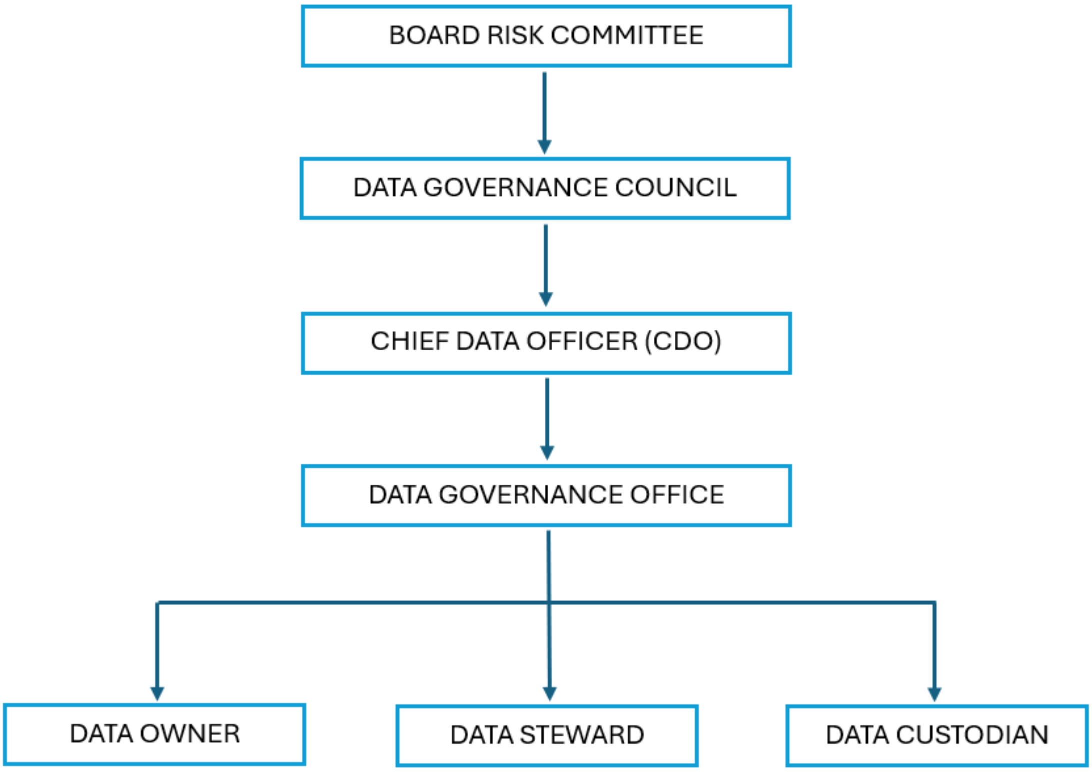
✓ Improve business process efficiency

✓ Enable reliable AI and analytics initiatives

✓ Establish accountability for data quality

3. DATA QUALITY GOVERNANCE MODEL

Governance Structure



Roles & Responsibilities

Data Governance Council

Responsibilities:

- Approve Data Quality Policies
- Monitor Enterprise Data Quality KPIs
- Escalate Critical Data Quality Issues
- Review Regulatory Data Quality Performance

Chief Data Officer (CDO)

Responsibilities:

- Own Enterprise Data Quality Strategy
- Define Data Quality Standards
- Monitor Enterprise Data Quality Score
- Report to Executive Management

Data Owner

Responsibilities:

- Accountable for data quality within the domain
- Approve Data Quality Rules
- Resolve Critical Data Issues

Examples:

- Head of Retail Banking
- Head of Lending
- Head of Operations

Data Steward

Responsibilities:

- Monitor data quality
- Perform issue investigation
- Execute remediation activities
- Maintain business definitions

Data Custodian

Responsibilities:

- Implement technical controls
- Monitor system performance
- Support remediation activities

4. DATA QUALITY DIMENSIONS

The Digital Bank shall assess data quality across six standard dimensions.

4.1. Accuracy

Definition: Data correctly represents real-world values.

Example:

Data Element	**Valid Example**
Customer Name	John Smith
Date of Birth	15-Jan-1990
Loan Amount	IDR 500,000,000

Target: 98%

4.2. Completeness

Definition: All required data fields are populated.

Example:

Required Fields:

- NIK
- Customer Name
- Mobile Number
- KYC Status

Target: 95%

4.3. Consistency

Definition: Data values remain consistent across systems.

Example:

System	Customer Segment
CRM	Premium
Core Banking	Premium

Target: 98%

4.4. Timeliness

Definition: Data is available when needed.

Examples:

- Real-time transaction data
- Daily regulatory reporting

Target: 95%

4.5. Validity

Definition: Data conforms to business rules and formats.

Example:

Valid Email: customer@email.com

Invalid Email: customer@email

Target: 99%

4.6. Uniqueness

Definition: No duplicate records exist.

Example:

One CIF per customer.

Target: 99.5%

5. CRITICAL DATA ELEMENTS (CDEs)

The following data domains are designated as Critical Data Elements.

Customer Domain

Examples:

- CIF Number
- Customer Name
- NIK
- Passport Number
- Date of Birth
- Customer Risk Rating

Account Domain

Examples:

- Account Number
- Account Status
- Account Type
- Branch Code

Loan Domain

Examples:

- Loan ID
- Loan Amount
- Interest Rate
- Loan Status

Transaction Domain

Examples:

- Transaction ID
- Transaction Amount
- Transaction Date
- Transaction Type

Regulatory Domain

Examples:

- OJK Reporting Data
- BI Reporting Data
- AML Monitoring Data

6. DATA QUALITY RULE FRAMEWORK

Rule Categories:

6.1. Completeness Rules

Example: Customer NIK cannot be null.

6.2. Accuracy Rules

Example: Customer age must be ≥ 17 years.

6.3. Validity Rules

Example: Email format must comply with RFC standards.

6.4. Consistency Rules

Example: Customer segment must match across CRM and Core Banking.

6.5. Uniqueness Rules

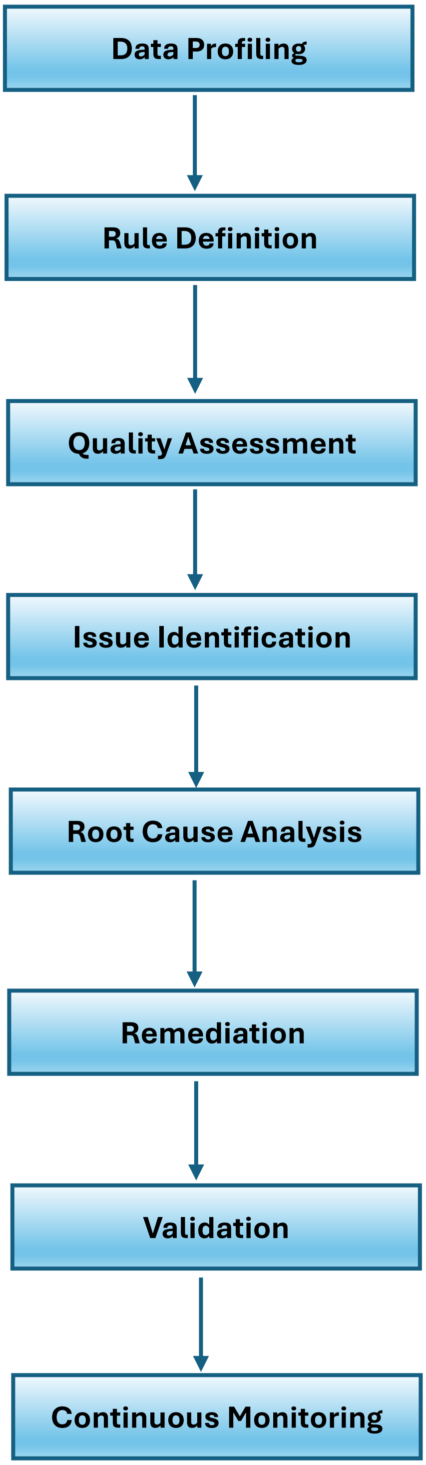
Example: One NIK may only belong to one CIF.

6.6. Timeliness Rules

Example: Transaction records must be loaded to Data Warehouse within 15 minutes.

7. DATA QUALITY ASSESSMENT PROCESS

End-to-End Process



8. DATA PROFILING FRAMEWORK

Data profiling shall be conducted before onboarding data into the governance program.

8.1. Profiling Activities

Structure Analysis
Examples:

- Data Type Validation
- Length Validation

Content Analysis
Examples:

- Null Values
- Duplicate Records
- Relationship Analysis

Relationship Analysis
Examples:

- Parent-Child Relationships
- Referential Integrity

9. DATA QUALITY MONITORING FRAMEWORK

9.1. Monitoring Frequency

Data Domain	Frequency
Customer Data	Daily
Transaction Data	Real-Time
Account Data	Daily
Loan Data	Daily
Regulatory Data	Daily
Metadata	Weekly

9.2. Monitoring Channels

- Data Quality Dashboard
- Data Catalog
- Data Governance Portal
- Automated Alerts

10. DATA QUALITY ISSUE MANAGEMENT

10.1. Severity Classification

Severity	Description
Critical	Regulatory or customer impact
High	Significant operational impact
Medium	Moderate business impact
Low	Minor issue

10.2. Escalation Path

Data Steward
↓
Data Owner
↓
Data Governance Office
↓
Chief Data Officer
↓
Data Governance Council

11. ROOT CAUSE ANALYSIS (RCA)

Mandatory for:

- Critical Issues
- High Issues
- Repeat Issues

Methods:

- 5 Whys
- Fishbone Analysis
- Process Analysis
- System Analysis

12. DATA QUALITY KPI FRAMEWORK

Enterprise KPIs

KPI	Target
Enterprise Data Quality Score	>95%
Accuracy Score	>98%
Completeness Score	>95%
Consistency Score	>98%
Validity Score	>99%
Uniqueness Score	>99.5%
Data Ownership Coverage	100%
Critical Issues Resolved Within SLA	≥95%
Duplicate Customer Rate	<0.5%
Regulatory Reporting Accuracy	100%

13. DATA QUALITY DASHBOARD (Executive View)

Metric	Target	Status
Enterprise DQ Score	>95%	🟢
Customer Data Quality	>95%	🟢
Transaction Data Quality	>98%	🟢
Regulatory Data Quality	100%	🟢
Open Critical Issues	0	🟢
Duplicate Customer Rate	<0.5%	🟢

14. TARGET STATE DATA QUALITY ARCHITECTURE

SOURCE SYSTEMS

Core Banking
CRM
Mobile Banking
Loan System
AML System
Payment System

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DATA QUALITY LAYER

Data Profiling
Data Validation
Business Rules Engine
Data Monitoring
Issue Management

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MDM PLATFORM

Customer Master
Product Master
Vendor Master

↓

ENTERPRISE DATA PLATFORM

Data Warehouse
Regulatory Reporting
BI & Analytics
AI/ML Platform

15. DATA QUALITY MATURITY MODEL

Level	Description
Level 1	Ad Hoc
Level 2	Managed
Level 3	Defined
Level 4	Measured
Level 5	Optimized

Current State (Example)
Level 2 – Managed

Target State
Level 4 – Measured

Long-Term Vision
Level 5 – Optimized

16. SUCCESS CRITERIA

The Data Quality Management Framework will be considered successful when:

✓ Enterprise Data Quality Score exceeds 95%
✓ All Critical Data Elements have assigned owners
✓ Duplicate customer records are below 0.5%
✓ Regulatory reporting accuracy reaches 100%
✓ Critical data quality incidents are reduced by at least 20% year-over-year
✓ SLA compliance exceeds 95%
✓ Data quality monitoring is fully automated